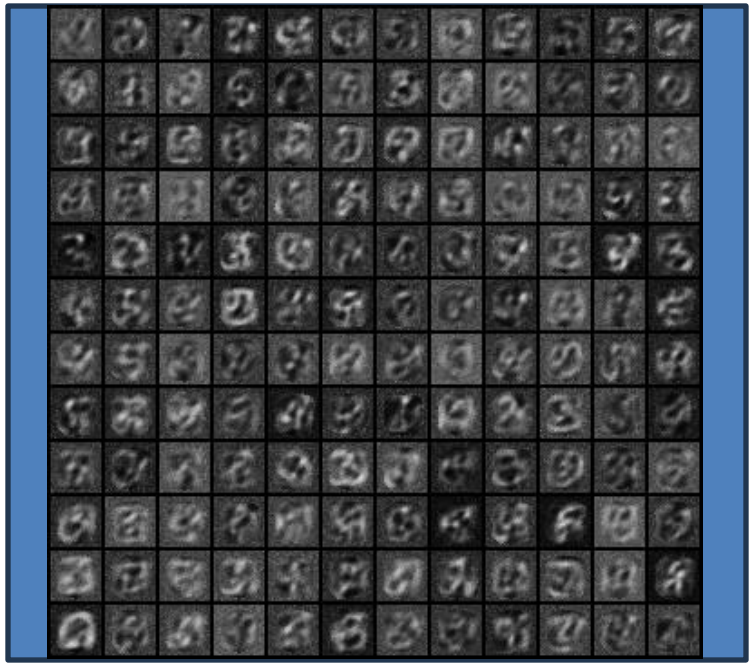
Assignment 4 Writeup

DO NOT TAG

Variational Autoencoder (VAE) Q1 - MNIST

1. Tune beta value with L2 reconstruction loss.
2. Keeping beta fixed switch reconstruction loss to L1.
3. Show examples of L1 and L2 recon loss:

L2:

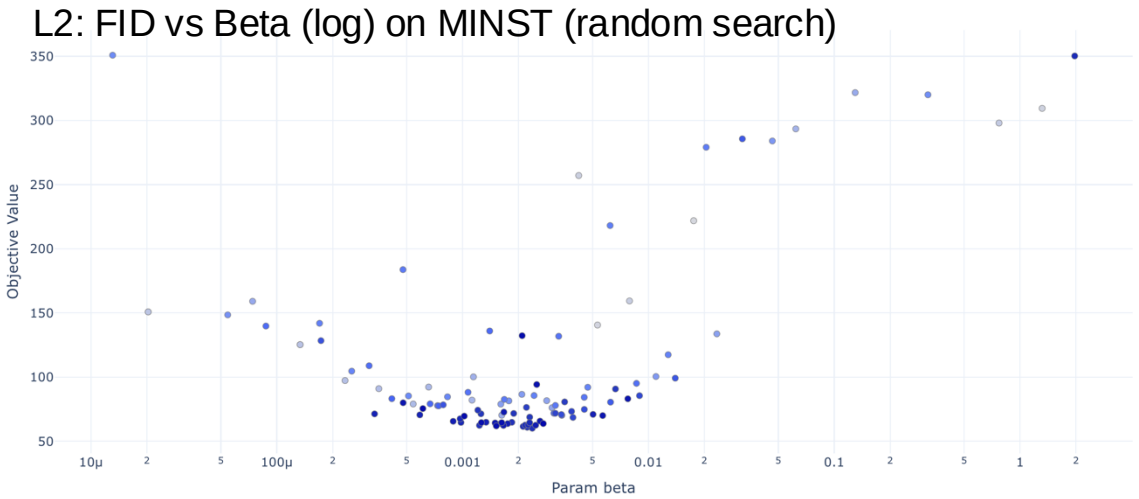


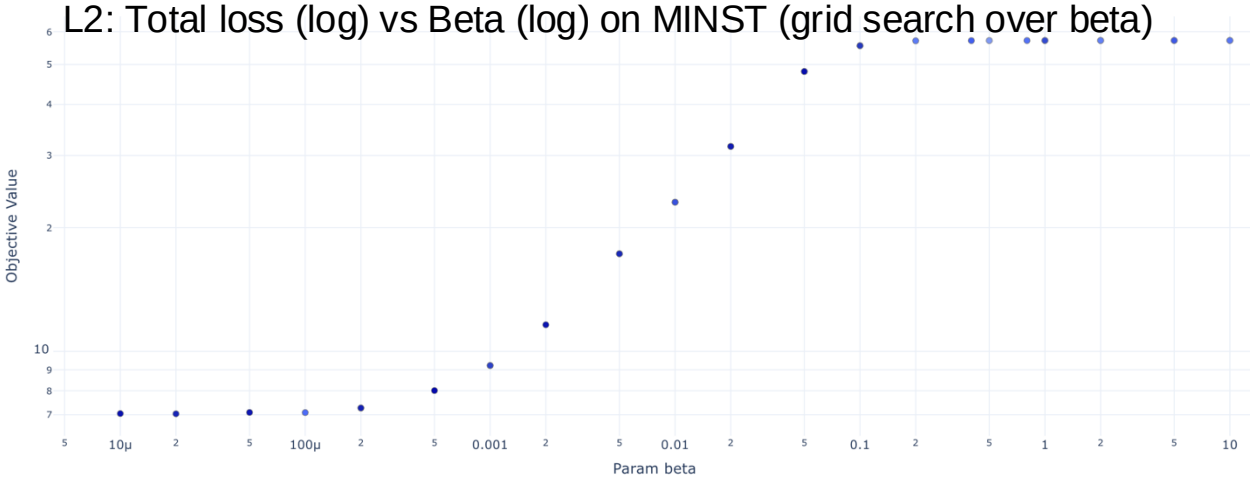
L1:



Variational Autoencoder (VAE) Q1 cont.

The best (lowest) FID score happened when the reconstruction loss was balanced with the KL loss. In the log-log plot of total loss vs. beta value on the previous slide values of beta < 0.0005 caused the reconstruction loss to dominate, and values of beta > 0.1 caused the KL loss to dominate. The value that that struck the best balance was around on 0.002, the log scale makes the seem a little low, but helps with the visualization of the rest of the range.





Variational Autoencoder (VAE) Q1 cont.

Explain your observations for VAE with L1 vs L2?

The images created with L2 loss are smoother and more coherent, with less noise and better preservation of the overall digit structure. However, the images tend to appear blurred, with soft edges less fine detail. In contrast, images with L1 loss appear sharper and more defined, especially around the edges of digits. L1 loss preserves high-intensity pixel differences better, which results in clearer strokes but also introduces more noise and visual artifacts, like fuzziness/patchiness or partial digits. This is because L1 encourages sparsity and penalizes large errors less aggressively than L2, leading to images that emphasize strong features at the cost of consistency. Overall, L2 is better for producing smooth and structurally accurate images, while L1 is better for highlighting fine details and producing sharper images, although sometimes at the cost of completeness.

Variational Autoencoder (VAE) Q2 - MNIST

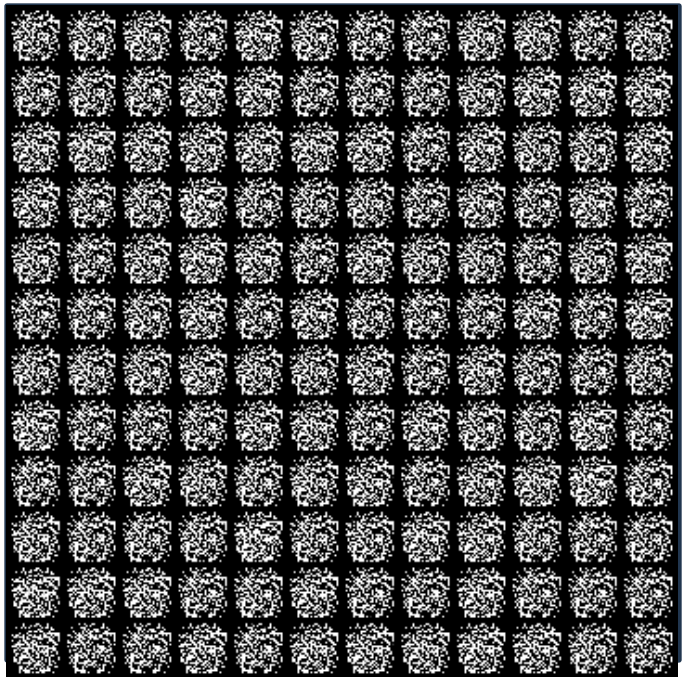
As we optimize for our few-shot setting (KNN classifier) what happens to the quality of generated samples? Share an example and explain your observations.

Optimizing for a few-shot KNN is a slightly different task than trying to create the best general generative VAE model. This inductive bias KNN creates for classification encourages the latent space to focus on reconstructing features that are distinguishable between the different classes. A higher KL divergence can be helpful for this but can make the reconstructed images look worse or at least blurrier as seen here.



Generative Adversarial Network - MNIST

There is a common failure mode of GAN training termed *mode collapse*. that is when the generator learns to fool the discriminator producing only a certain subset of P(x). Find hyperparameters that exhibit this failure mode and show generated examples.



```
data:
  dataset: mnist # mnist or fashionmnist
train:
  batch_size: 32 # Small batch size for instability
  lr: 0.01 # High learning rate
  n_epochs: 50
  num_workers: 4
optimizer:
  type: 'adamw'
  weight_decay: 0.01 # High weight decay to weaken discriminator
network:
  model: gan
  hidden_dim: 128 # Small hidden dimension
  latent_dim: 4 # Very small latent dimension to restrict diversity
gan:
  leaky: false # Regular ReLU for weaker discriminator gradients
```

Simple Diffusion

Insert your simple diffusion visualizations here:

iteration 0

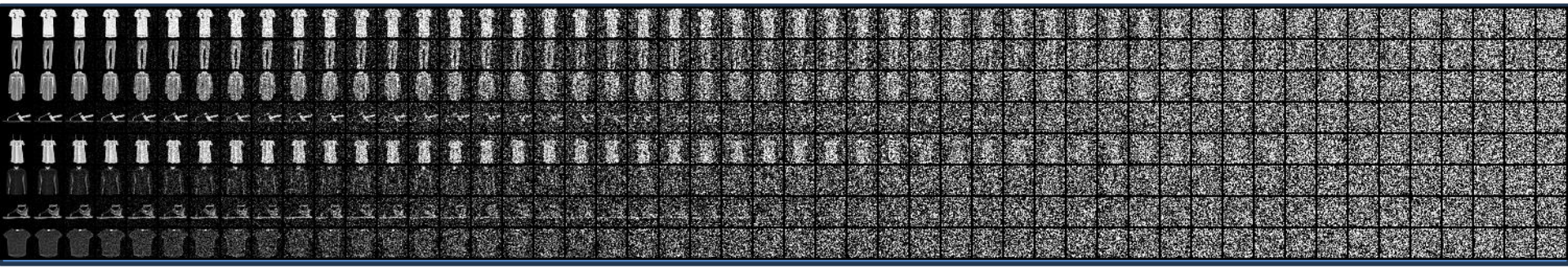


iteration 99



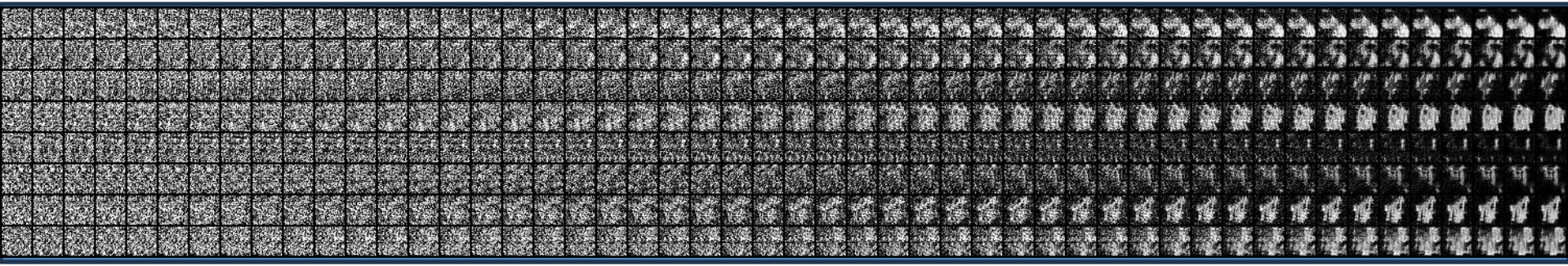
DDPM - MNIST

Insert your forward diffusion visual here (any epoch is fine):

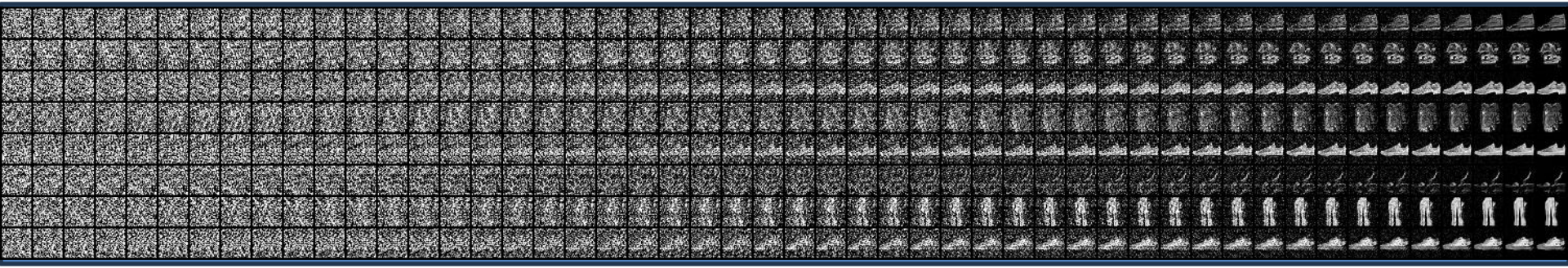


DDPM - MNIST

After tuning only noise schedule and timesteps:
Insert your reverse diffusion visual here **(epoch 0)**:



Insert your reverse diffusion visual here **(epoch 9)**:



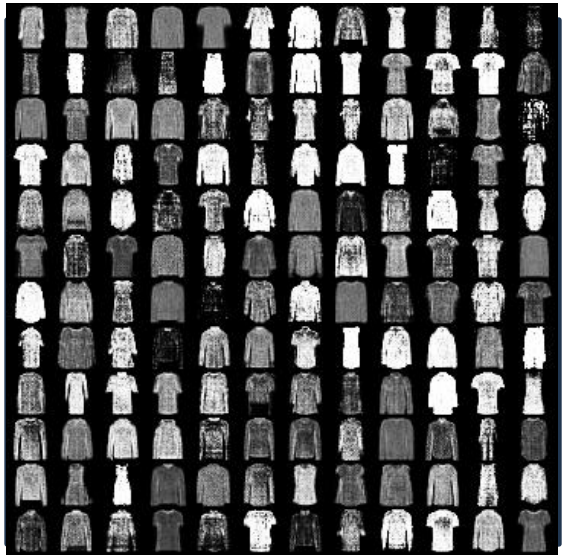
Tune each model (fashionMNIST) to achieve the best images and show your results:

VAE



FID: 76.12

GAN



FID: 114.42

Diffusion



FID: 50.11

Compare and contrast the results from the different models. Why might we see the results that we see? what role does the target dataset play?

The comparison shows distinct visual outcomes and FID scores for the three models on FashionMNIST. The VAE produces blurry images (FID: 76.12) due to its optimization objective, which tends to average pixel values. The GAN generates sharper images but suffers from artifacts some mode collapse (FID: 114.42), likely due to instabilities inherent in adversarial training. The Diffusion model achieves the best results, producing the clearest, most realistic images with the lowest FID score (50.11), reflecting its ability to capture fine details and global coherence through its iterative denoising process.

The FashionMNIST dataset, while more complicated than MNIST, is still relatively simple. It highlights the differences between models. The VAE's blurriness, the GAN's artifacts, and the Diffusion model's high fidelity are all clearly visible. The well-defined nature of the real data distribution in FashionMNIST lets the FID score to capture how well each model replicates it, confirming the visual assessment that the Diffusion model performs best in this comparison.

FID is commonly used to benchmark generative models. Is it a good metric? Why or why not?

Fréchet Inception Distance (FID) is a decent benchmark, although it has some tradeoffs and limitations. FID uses a pretrained Inception-V3 model to generate embeddings for real images and generated images and then compare the distance between them. The embeddings capture high-level semantic information and which can correlate with perceived image quality. FID also penalizes mode collapse because the generated distribution of features gets narrower. This approach allows for a comparison between different models and dataset, allowing for a standard benchmark to measure the state of the art (SOTA).

Some drawbacks are the reliance on Inception-V3 that was pretrained on ImageNet. More accurate classifiers are available and testing on images that are out of distribution from ImageNet can cause issues. The FID score doesn't always align with human judgement. FID is also sensitive to small sample sizes.

Overall FID is a useful benchmark but is far from a perfect proxy for human judgement.